Enumerative combinatorics

For a finite set A, we denote:

1) $|A|$ - the number of elements (the cardinality of A)

2) $A \times B = \{(a,b) : a \in A, b \in B\}$ - Cartesian product

3) $A \cup B$ - union

4) $A \cap B$ - intersection

5) $A \setminus B$ - difference

1. Counting by bijections

Def. 1) Let $f: A \to B$ be a function:

- f is injective if different elements have different images: $x \neq y \Rightarrow f(x) \neq f(y)$

- f is surjective if every element of B is the image of an element of A.

- f is bijective if it is injective and surjective, that is, if every element of B is the image of exactly one element of A.

Obs. A function $f: A \to B$ is bijective $\Leftrightarrow$ f is invertible, that is, there is $g: B \to A$ with $g(f(x)) = x$, $\forall x \in A$ and $f(g(y)) = y$, $\forall y \in B$.

Def. 2) A set A has n elements, $|A|=n$, if there is a bijection $f: \{1,2,\dots,n\} \to A$. The empty set has 0 elements.

Th. 3) (The bijection principle) Let A and B be two finite sets:

a) $|A|=|B| \iff$ there is a bijection $f: A \to B$

b) $|A| \leq |B| \iff$ there is an injective function $f: A \to B$

Method. When a set is hard to count directly, we look for a bijection between it and a set that is easy to count. To show that a correspondance is bijective, we also describe the inverse procedure ("the way back").

ex.: $3, 10, 17, 24, \dots, 2026$ /4

$7, 14, 21, 28, \dots, 2030$ /:7

$1, 2, 3, 4, \dots, 290 \Rightarrow 290$ nr

$100 \cdot 7 - 4 = 700 - 4 = 696$

$f(a) = \frac{(a+4)}{7}$ $f: \{3, 10, \dots, 2026\} \to \{1, 2, \dots, 290\}$

Obs. (Watch out for +1). An arithmetic progression $a, a+r, a+2r, \dots, b$ has $\frac{b-a}{r}+1$ terms.

ex.: $3^0 = 1$
$3^1 = 3$
$3^0 + 3^1 = 4$

$2^0 = 1$
$2^1 = 2$
$2^0 + 2^1 = 3$

$3^2 = 9$
$3^0 + 3^2 = 10$
$3^1 + 3^2 = 12$
$3^0 + 3^1 + 3^2 = 13$

$\Big| \longrightarrow$

$2^2 = 4$
$2^0 + 2^2 = 5$
$2^1 + 2^2 = 6$
$2^0 + 2^1 + 2^2 = 7$

$202600_{(10)} = 1010\,2122\,0201_{(3)}$

202600	3
67533	1
22511	0
7503	2
2501	0
833	2
277	2
92	1
30	2
10	0
3	1
1	0
0	1

$\Rightarrow 10101111111111_{(3)} =$

$= 200110_{(10)}$

$= 2^{11} + 2^9 + 2^7 + 2^6 + \dots + 2^0$

$=$ 2815 numere

$2026_{(10)} = 11111101010_{(2)} = 3812_{(10)}$

2. The product rule (the rule of "and")

ex. A → B 6 routes
B → C 5 routes
$\Big| \Rightarrow 6 \cdot 5 = 30$ possibilities

Th. 1) (The product rule) For any finite sets $X_1, X_2, \dots, X_n$:

$$|X_1 \times X_2 \times \dots \times X_n| = |X_1| \cdot |X_2| \cdot \dots \cdot |X_n|$$

2) (The product rule, in words) Each object of a set M is built by a sequence of n choices, made one after the other, so that:

(i) at step i there are exactly m_i options, whatever the choices at the previous step.

(ii) each object of M is obtained from exactly one sequence of choices.

Then $|M| = m_1 \cdot m_2 \cdot \ldots \cdot m_n$

Ex.: $\underline{9}\ \underline{9}\ \underline{8} = 9 \cdot 9 \cdot 8 = 648$ de numere

3. Double counting and the division rule:

Ex. A, B, C, D, E, F, G, H

1 player -> 7 pos.
2 player -> 5 pos
3 player -> 3 pos
1 player -> 1 pos

$\Rightarrow 7 \cdot 5 \cdot 3 \cdot 1 = 105$

A, B, C, D
1 player -> 3 pos
1 player -> 1 pos

$\Rightarrow 3 \cdot 1 = 3$

Obs. Where there is no order, we impose one.

Th. 1) (The division rule) If $f: A \to B$ and every element of B is the image of exactly $k \geq 1$ elements of A, then $|A| = k|B| \Rightarrow |B| = \frac{|A|}{k}$.

Obs. 1) A set with n elements has 2^n subsets

2) $2m$ people can be split into pairs in:

$$\frac{(2m)!}{2^m \cdot m!} = (2m-1)!! = (2m-1)(2m-3)\ldots 3 \cdot 1 \text{ ways}$$